

Main Body

Contracted Demand: 4,600 kVA

Penalty Trigger: Exceeding 110% of contracted demand

Penalty Rate: ₹425 per excess kVA per billing month

Measurement Basis: 15-minute rolling peak average

Demand Threshold Calculation:

$$4,600 \times 110\% = 5,060 \text{ kVA}$$

Operational Notes

- Shift overlap period: 45–90 minutes during handover.
 - Prototype testing consumes high instantaneous load.
 - Demand peaks are not averaged across hours.
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Appendix A (Small Font Clause)

Prototype testing line load:

- Included in measured plant peak.
 - Excluded from contracted demand eligibility basis.
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Appendix B

Production reporting policy:

- Production_Units includes rejected units.
 - Net saleable units require manual exclusion of rejected count.
 - Energy intensity dashboard uses gross production count.
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Determinism Check

Correct solution must:

- Remove rejected units.
- Remove prototype energy from numerator.
- Adjust HVAC double-count months.
- Reconstruct idle compressor energy from specs.
- Apply demand eligibility clause correctly.
- Identify scheduling as root cause.

Golden outputs:

Corrected Energy Intensity $\approx 2,445\text{--}2,465$ kWh/unit

Avoidable Annual Cost $\approx ₹1.9\text{--}2.2$ crore

Root Cause = Operational Scheduling

Capex Decision = Defer furnace, implement scheduling reform